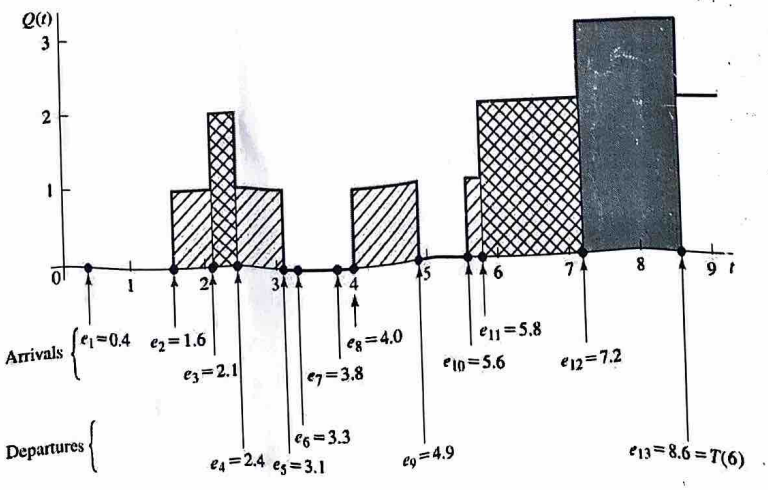
Modelling and Simulation BICT4702 Test 1

Time Allowed: 1hr 30Min

***Show your working clearly*

1. Differentiate the following types of models:
   a. *Deterministic vs Stochastic* [4 Marks]
   b. *Monte Carlo vs Dynamic* [4 Marks]
2. Describe the following systems in terms of their entities, activities, attributes, events, state variables.
   a. An interchange in Lilongwe that has experienced severe congestion.[4 Marks]
   b. computer system that needs faster processing speeds. [4 Marks]
3. During the lifecycle of a simulation, in the *model conceptualization* phase, the principle that modellers follow is to "start with a simple model and build towards greater complexity". With this in mind, assume you were conducting a simulation of a major traffic intersection in Lilongwe, with the objective of improving the current traffic flow. Provide three iterations, in increasing order of complexity, for this simulation study. [9 Marks]
4. The figure below is a plot of the number of customers in the que $(Q(t))$ against time t. Arrivals occur at times 0.4, 1.6, 2.1, 3.8, 4.0, 5.6, 5.8, and 7.2. Service completions occur at times 2.4, 3.1, 3.3, 4.9, and 8.6, and the simulation ends at time $T(6) = 8.6$.



a. Let $L(t)$ be the total number of customers in the system (those in service + those in the que) . Draw a plot of $L(t)$ vs t. [7 Marks]

b. From your plot in part (a), compute $\hat{l}(6)$ = the time-average number of customers in the system during the time interval $[0, T(6)]$. [3 Marks]

c. Let W_i = the total time in the system of the ith customer to finish service, which includes the time in queue plus the time in service of this customer. Compute $\hat{w}(m)$ = the average time in the system of the first m customers to exit the system, for m = 5. [3 Marks]

5. The time between arrivals at a drive through window of a fast food restaurant follows the distribution below. The service time distribution is also given in the table. Use the random numbers provided to simulate the activities of the first ten arrivals. Assume that the window opens at 11:00 am and that the first arrival is after this based on the first inter-arrival time.

Time between arrivals	*Probability*	*Service time*	*Probability*
1	0.2	1	0.3
2	0.3	2	0.5
3	0.3	3	0.2
4	0.2		

a. Construct a simulation table for the scenario above using the following random numbers: Random numbers for arrivals: 14, 74, 27, 03 Random numbers for service times: 88, 32, 36. [12 Marks]